

Nuclear Structure Investigations of Light Nuclei with the R3B Experiment

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R³B Experiment @FAIR

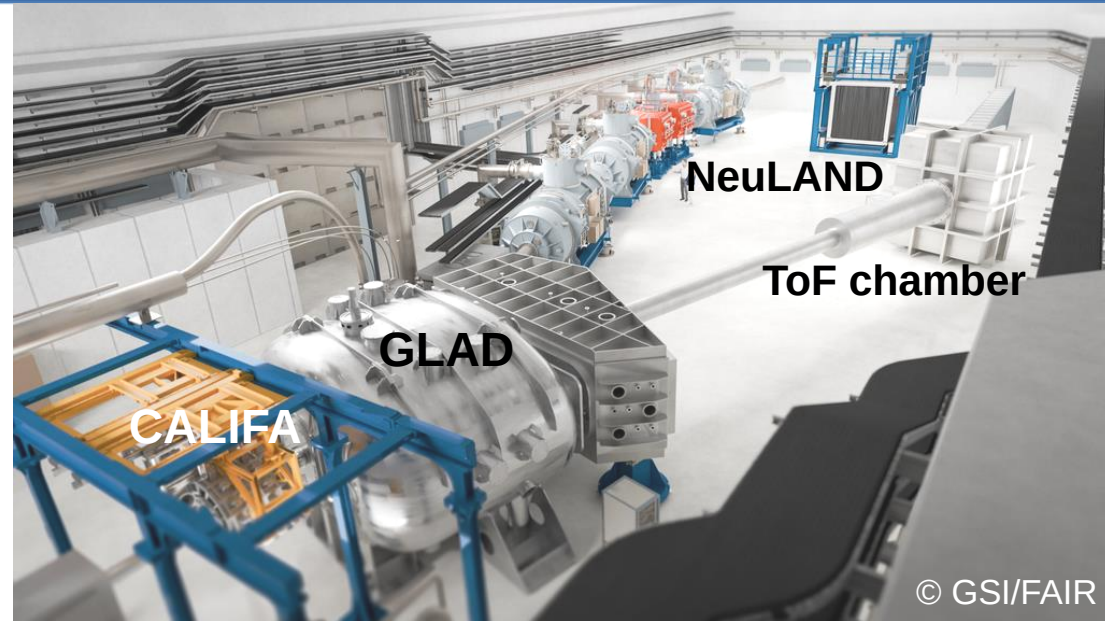
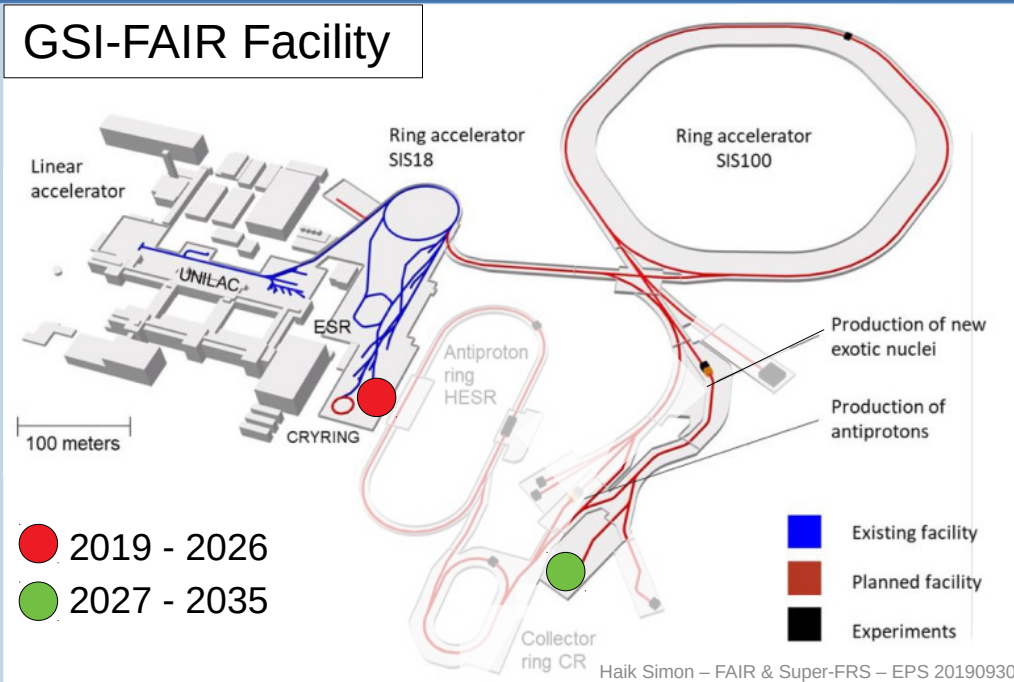
Total Interaction Cross Section $^{12}\text{C} + ^{12}\text{C}$

Quasi-Free Scattering $^{12}\text{C}(p,2p)^{11}\text{B}$

Fission Experiments @ R3B

ML for Cluster Reconstruction in CALIFA

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and the FAIR Phase-0 program

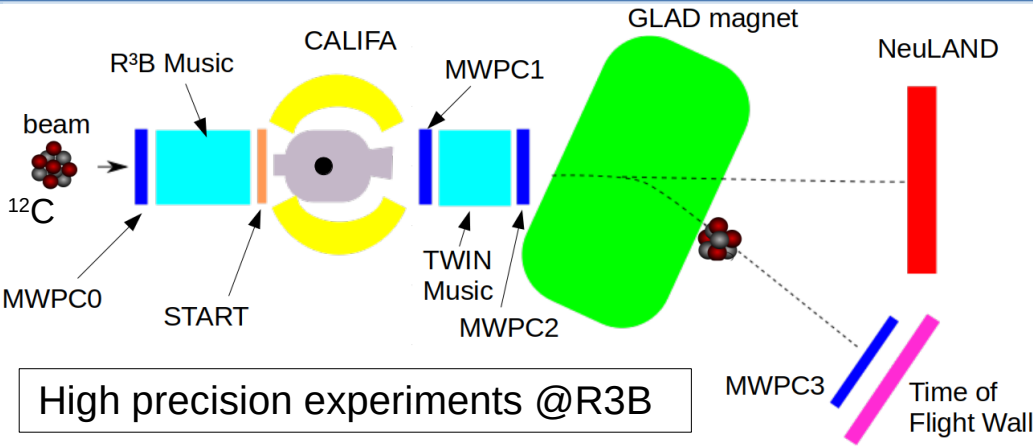


Experimental Approach:

- Inverse kinematics
- Kinematically complete measurements

Scientific goals:

- Study the structure of exotic, unstable nuclei
- Investigate nuclear fission dynamics and fission product properties
- Examine short-range correlations between nucleons



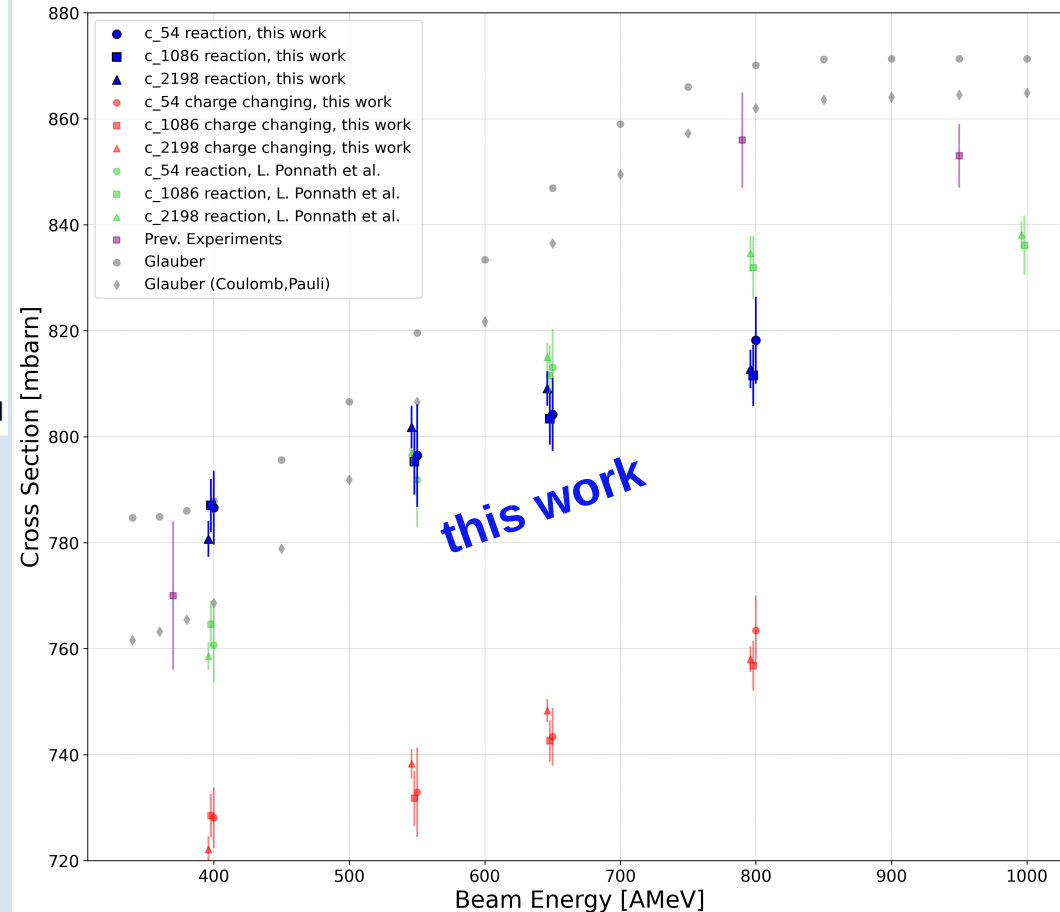
Motivation:

Quantify the uncertainty of the reaction model

Scientific goal:

Constrain Slope parameter of the symmetry energy of the EoS via precise σ_{dN} measurements

Measurement via transmission method:

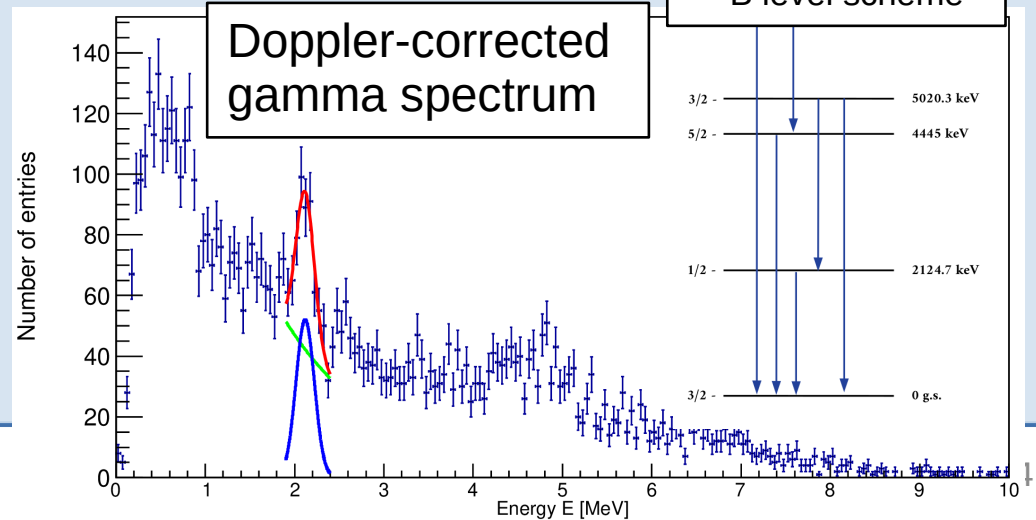
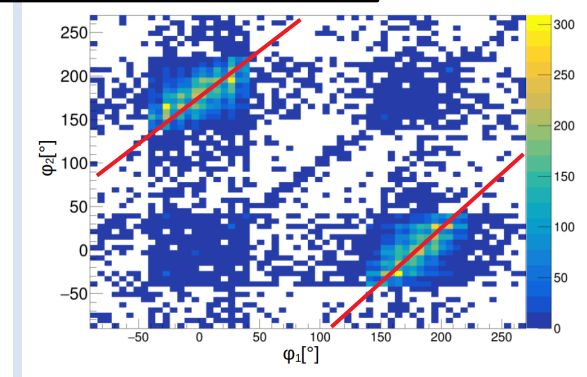
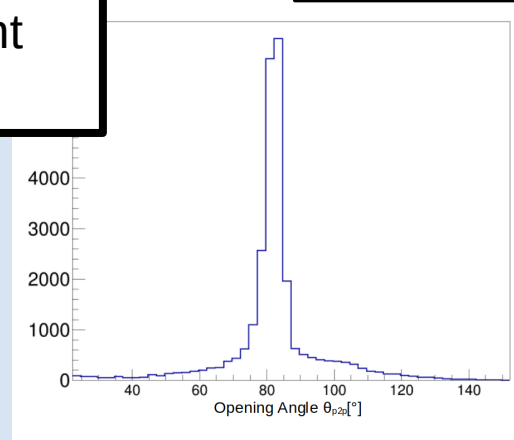
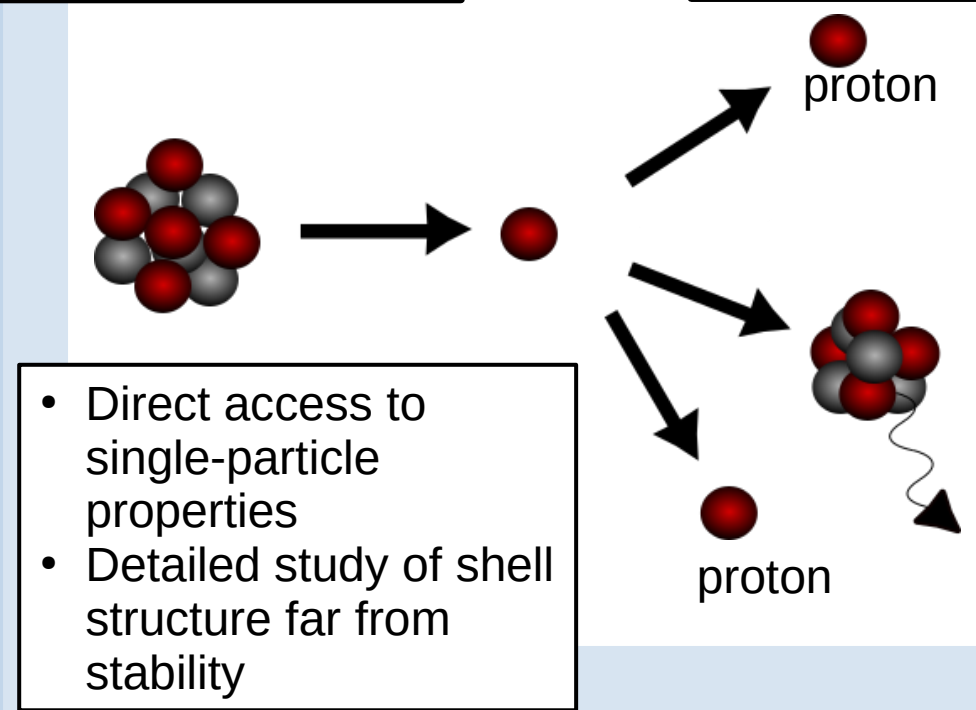


Quasi-Free Scattering $^{12}\text{C}(p,2p)^{11}\text{B}$

➤ ^{12}C beam
➤ proton like target

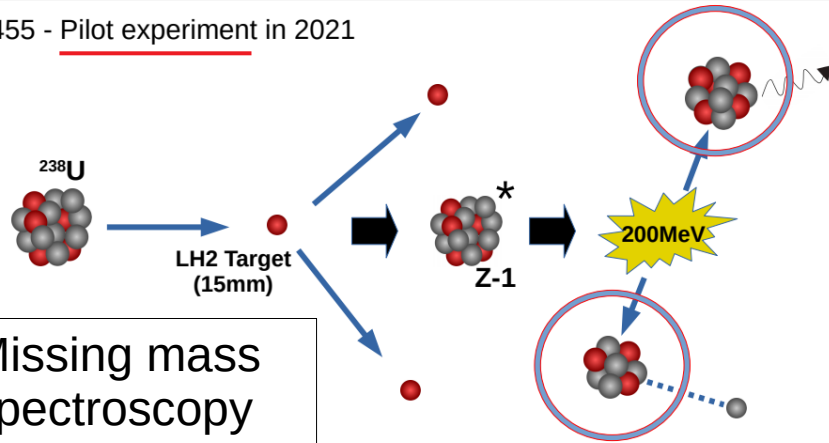
➤ 2 protons
➤ ^{11}B fragment (spectator)

two-proton correlations



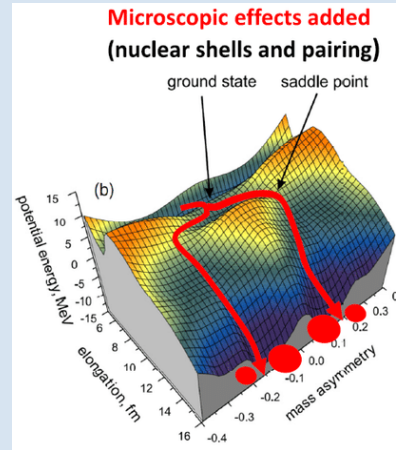
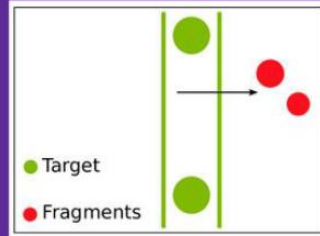
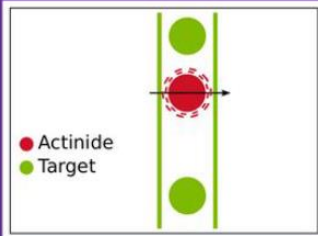
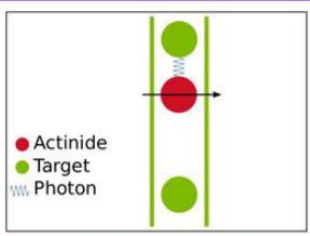
Fission via QFS: $^{238}\text{U}(p,2pf)$

S455 - Pilot experiment in 2021

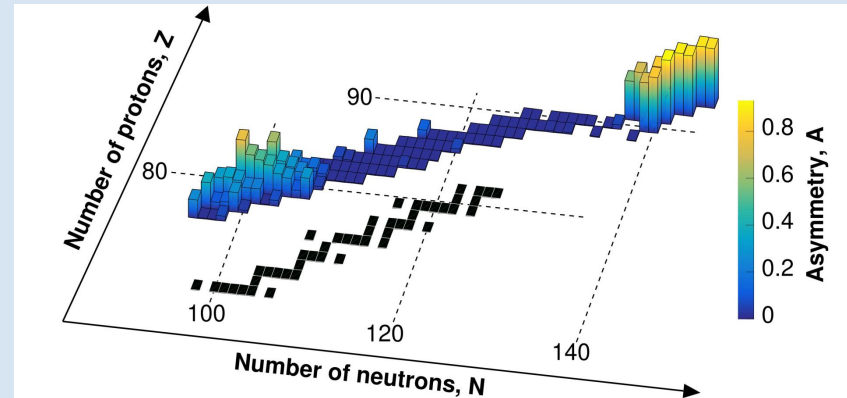
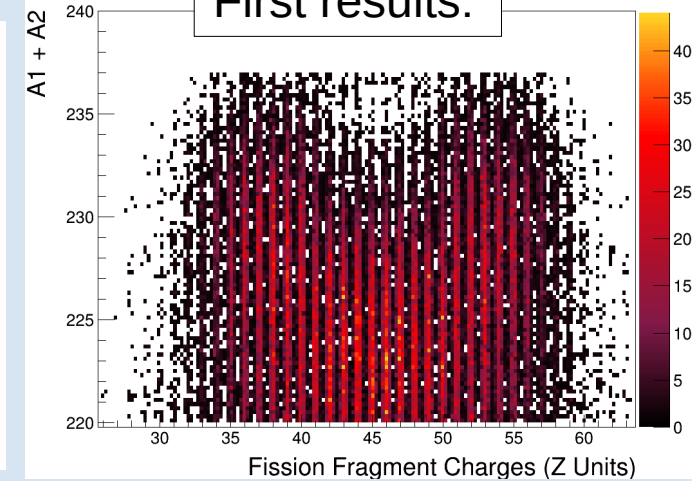


Missing mass spectroscopy

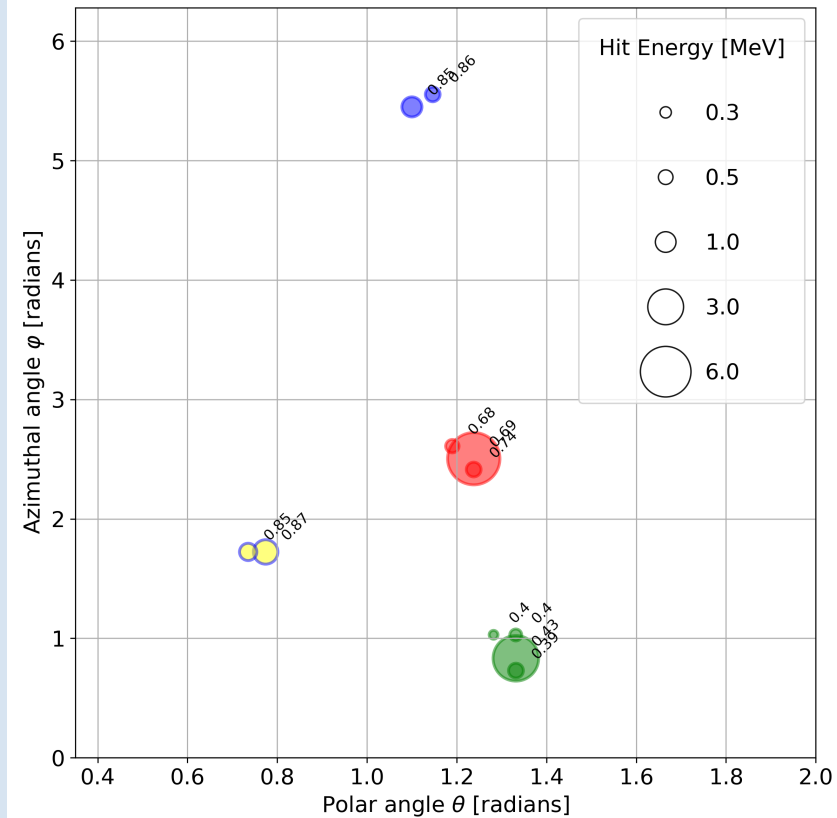
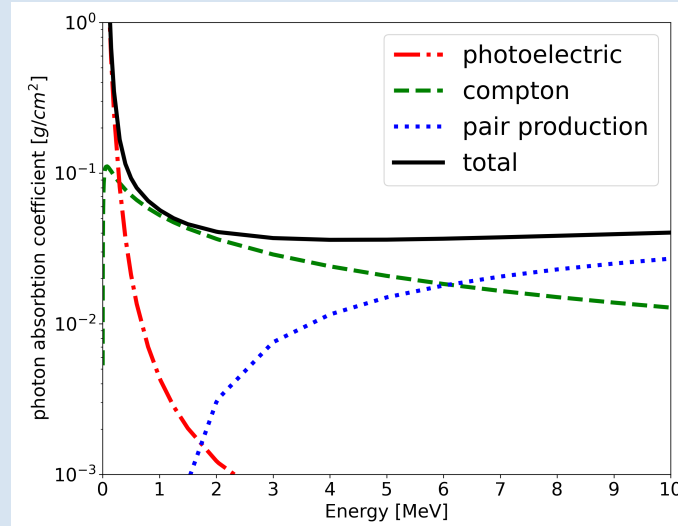
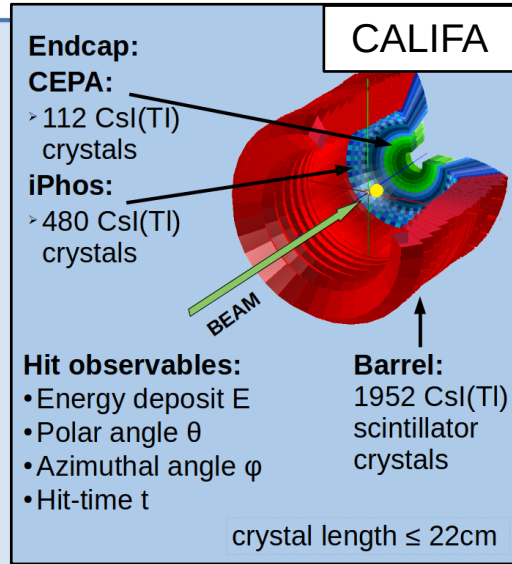
Complementary method via coulomb excitation



First results:



"An asymmetric fission island driven by shell effects in light fragments", Nature (2025)



Edge Detection Neural Network

2. Pairwise hit comparison (i, j)

1. Pre-clustering

